

GuardAi

Enhancing Robustness and Security of Edge AI Systems for Safety-Critical Applications

RAPTOR

Robustness Assessment of Patch Transfer and Operational Risk in Edge Object Detectors for Mobile Robots

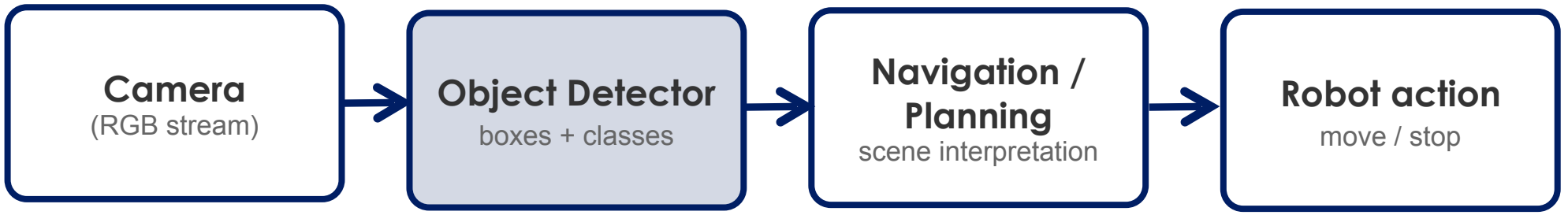
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ECCC

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Object Detection Drives Robot Decisions



Safety
Incorrect robot actions

Edge constraints
Limited compute & energy

Perception Overload
False detections burden downstream modules

Object detector failures become robot decision failures

Beyond Conventional Robustness Evaluation

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Conventional Evaluation

- ✓ Detection Accuracy (mAP)
- ✓ Attack Success Rate
- ✓ Single Detector

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But is the system robust in deployment?

↓

Cross-Model Transfer

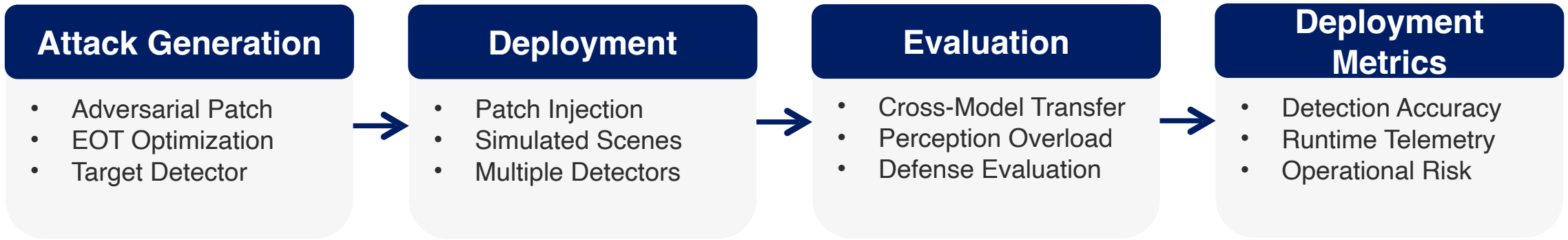
Perception Overload

Platform Telemetry

Defense Evaluation

RAPTOR evaluates robustness from a deployment perspective

RAPTOR Evaluation Framework



Experimental Setup

Attack Model

Adversarial Patch

White-box optimization

EOT optimization

Objective

Perception Overload

↑ False detections/image

↓ Detection accuracy

Evaluation

CARLA-GeAR Dataset

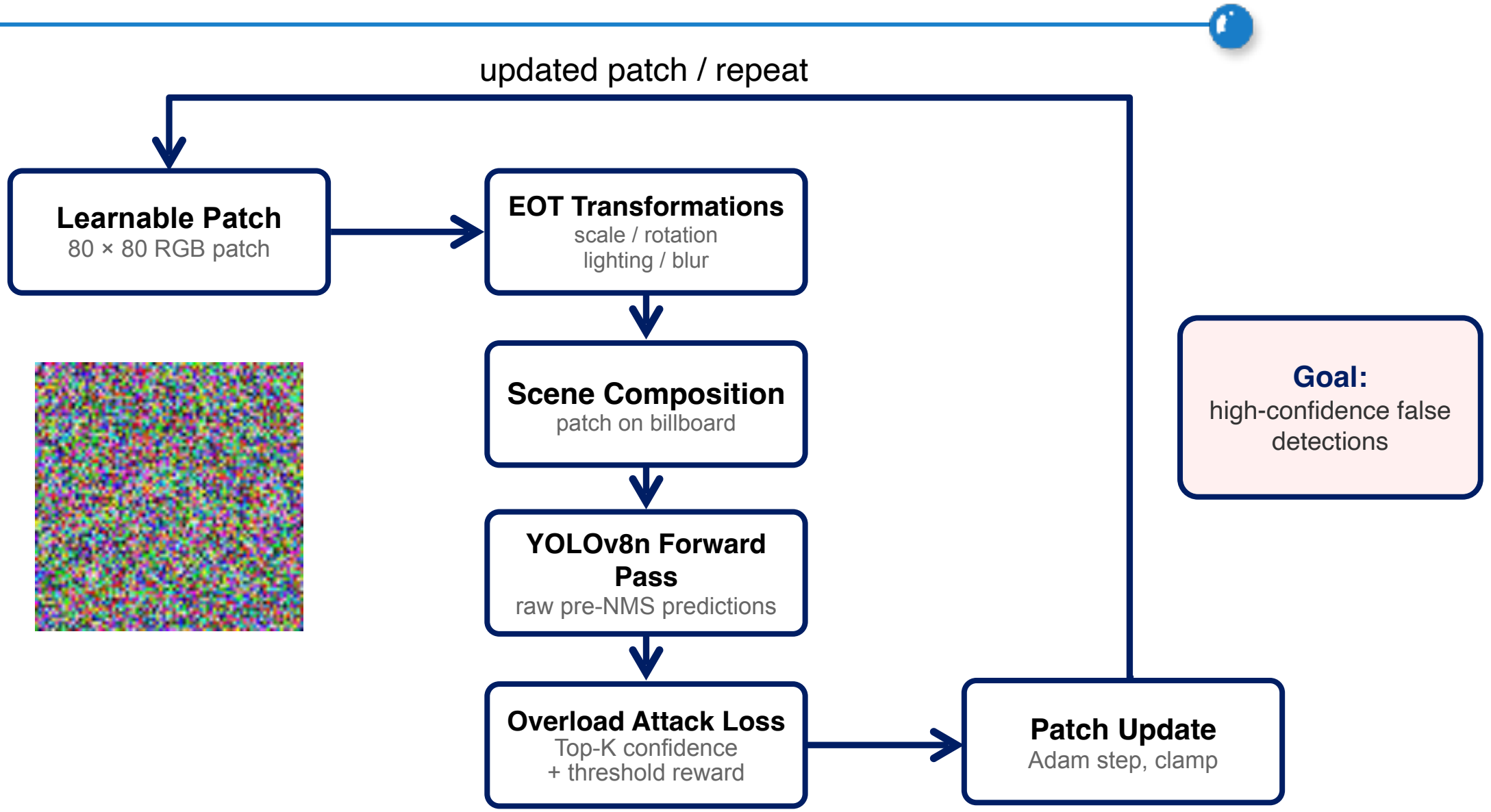
9 simulated billboard scenarios
5 detector configurations

Edge Deployment

Nova Carter - Jetson AGX Orin

Latency
Power
Energy/frame

Patch Optimization Pipeline



Attack Loss Setup: Pre-NMS Predictions

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The loss operates on raw detector outputs before Non-Maximum Suppression

Raw YOLOv8n predictions
pre-NMS detector outputs

↓

before filtering and NMS

↓

Class-agnostic confidence

$$s_i = \max_{c \in \{1,\ldots,C\}} p_{i,c}$$

Each raw prediction is scored by its strongest class probability.

Attack Loss: Optimizing for False-Positive Flooding

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Two complementary terms create false-positive flooding.

Confidence term

Depth: stronger predictions

$$\mathcal{L}_{\text{TopK}} = -\sum_{i \in \text{TopK}(s)} \log\left(s_i + \varepsilon\right)$$

K = 300

Pushes the strongest raw predictions toward high confidence.

Count term

Breadth: more predictions

$$\mathcal{L}_{\text{th}} = -\lambda \sum_{i=1}^{N} \max\left(0, s_i - \tau\right)$$

τ = 0.25, λ = 1.0

Rewards predictions that cross the confidence threshold.

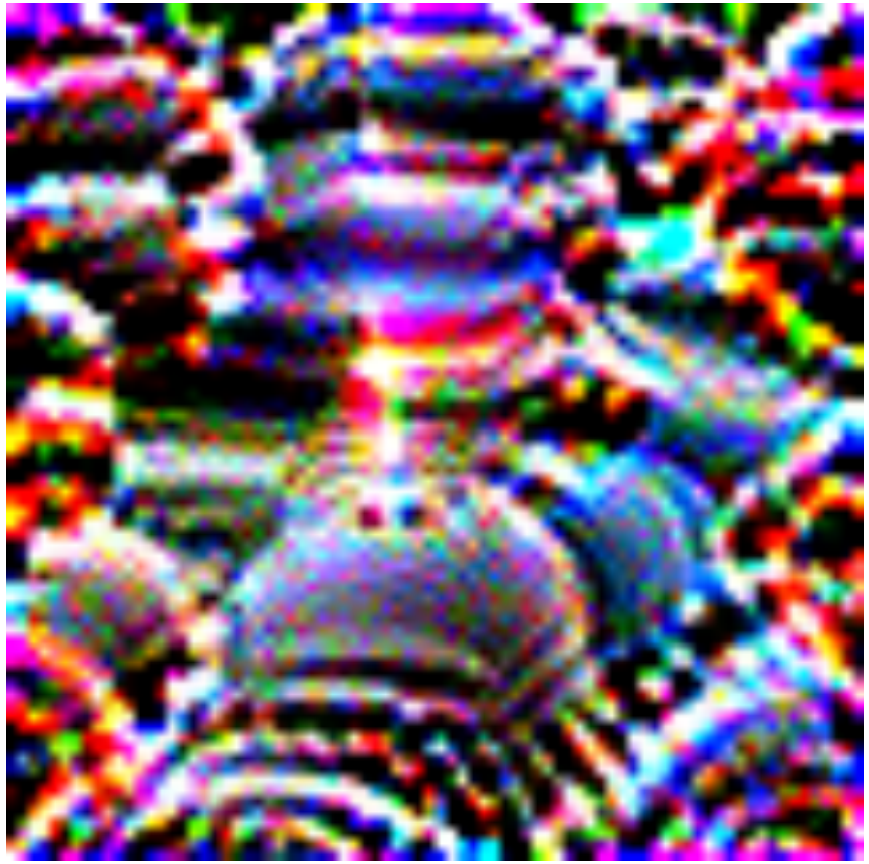
$$\mathcal{L} = \mathcal{L}_{\text{TopK}} + \mathcal{L}_{\text{th}}$$

confidence + count → false-positive flooding → perception overload

Generated Scenario-Specific Patches

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Each billboard scenario learns its own optimized patch.



Optimized for billboard01



Optimized for billboard02

Applying the Adversarial Patch

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Same scene before and after patch insertion



Clean Scene



Patched Scene

From Patch Insertion to Detector Response

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The patch creates extra detections around the billboard region

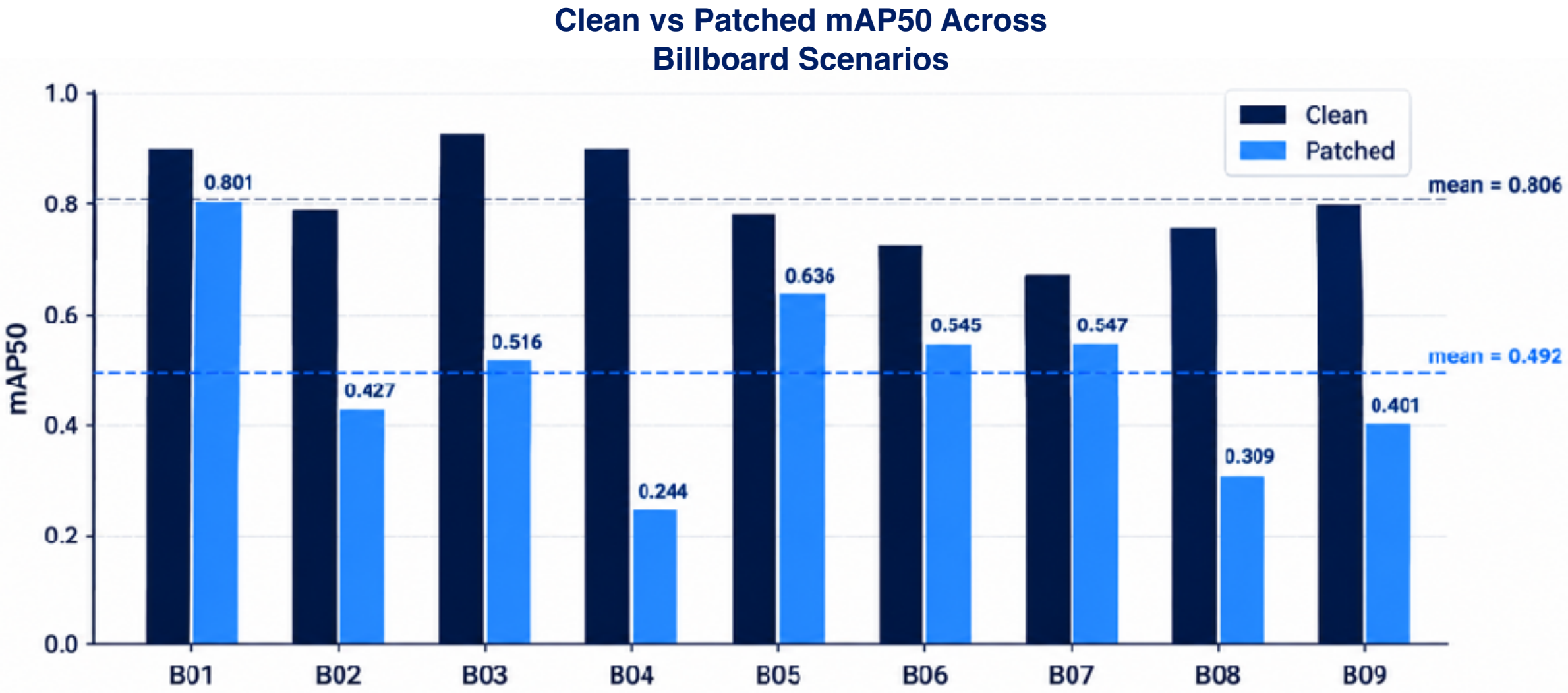


Clean Scene

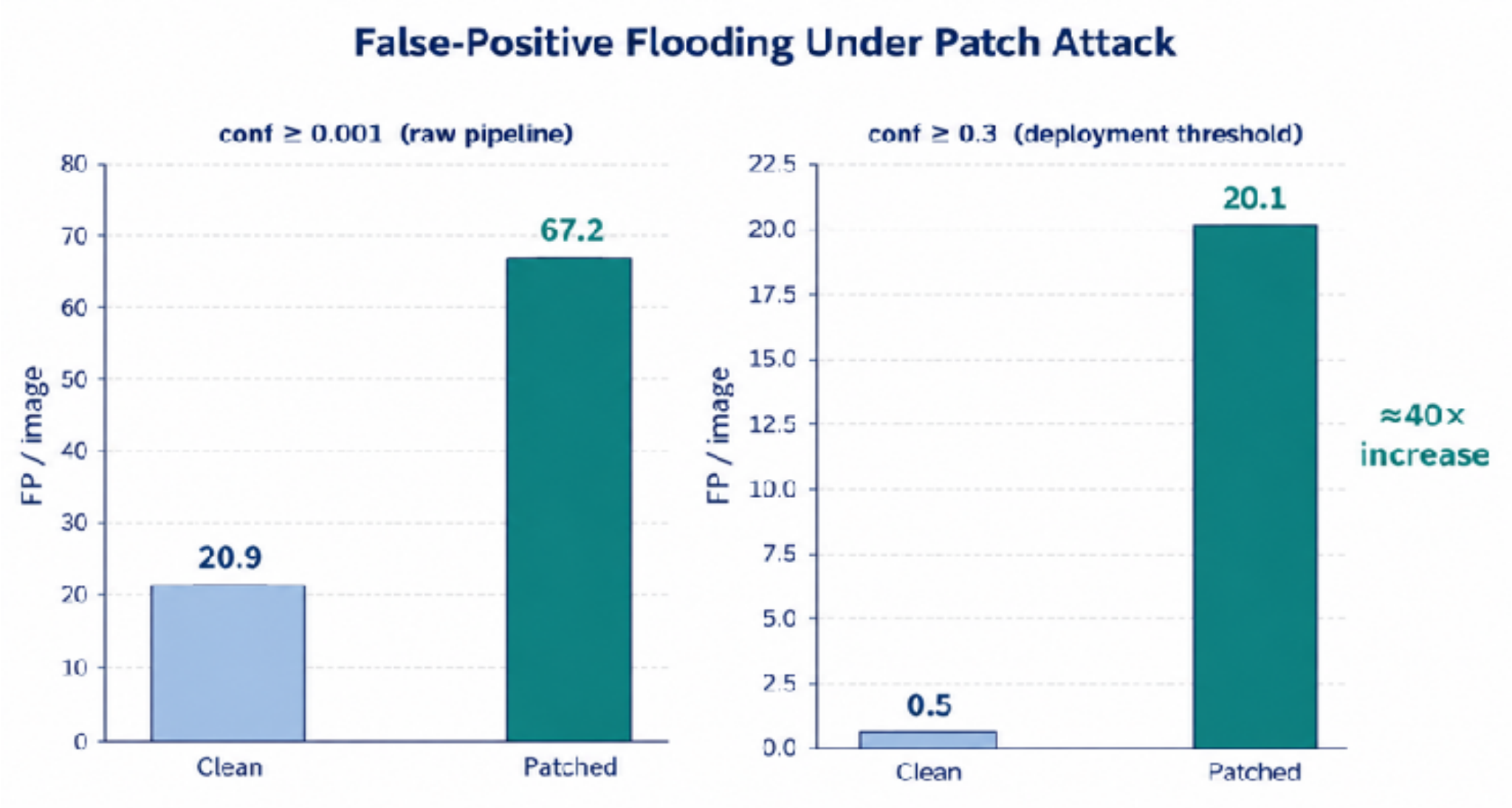


Patched Scene

Quantitative Attack Impact

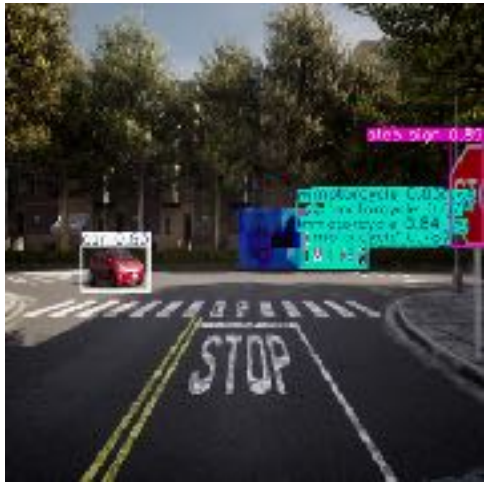


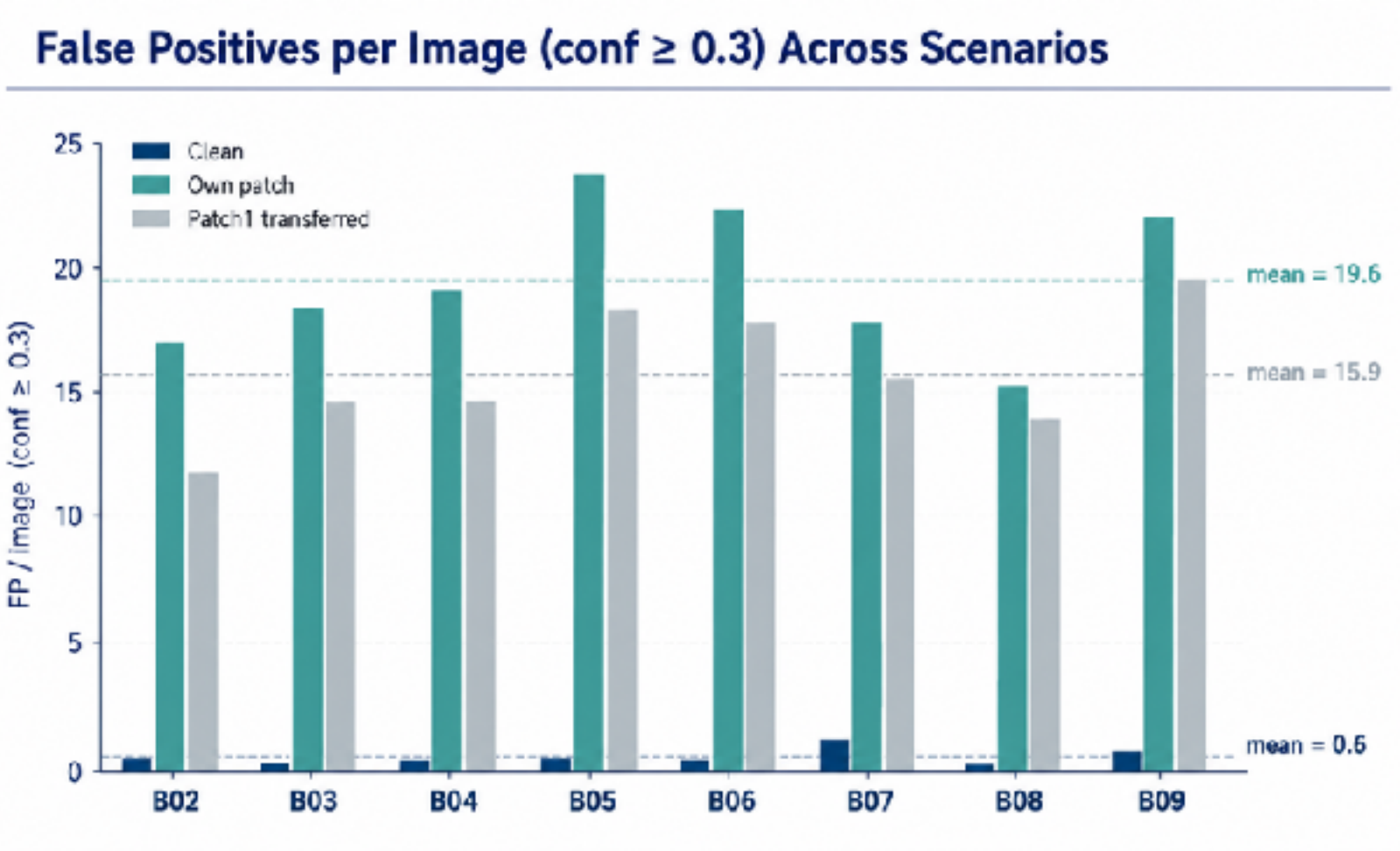
Quantitative Attack Impact



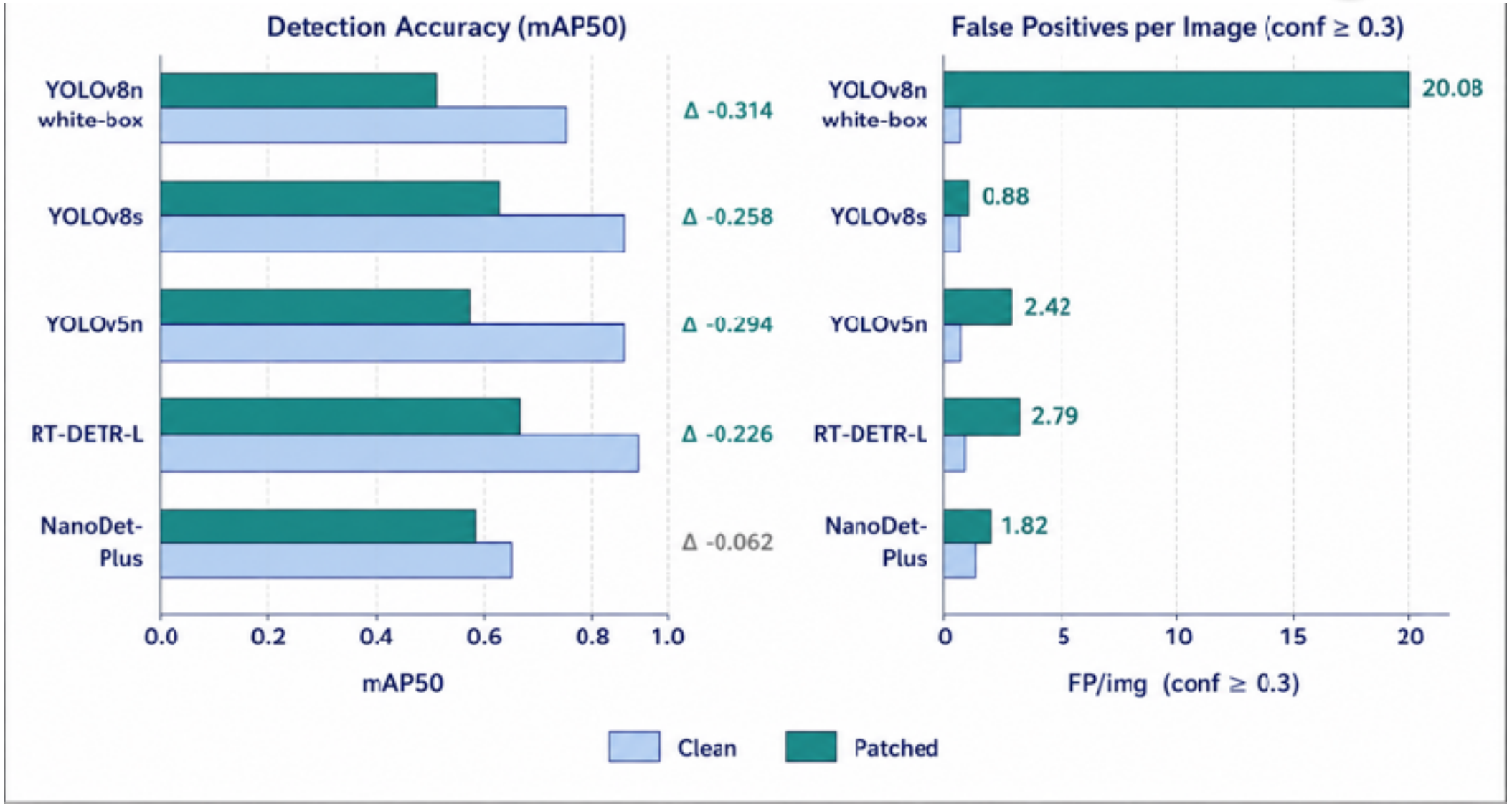
Cross-Scenario Transfer of Patch 1





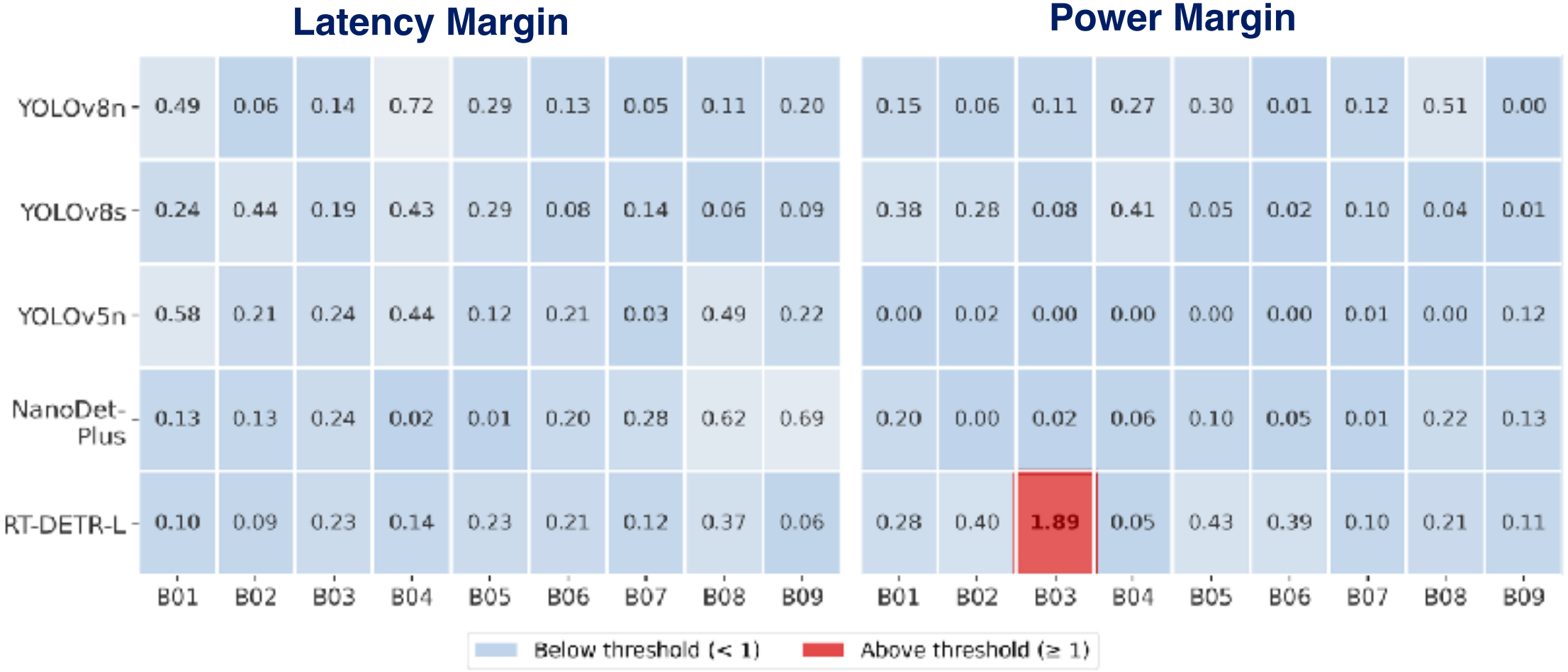


Cross-Architecture Transfer



Hardware Profiling

Hardware-invisible: latency and power remain within normal execution variability.



44 / 45 cases hardware-invisible

Defense Methods

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Whole-image baselines

JPEG compression

Bit-depth reduction

Gaussian blur

Median filtering

Whole-image transforms

Spatially selective defence

LGS

Local Gradient Smoothing

High-frequency map

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Noisy-pixel mask

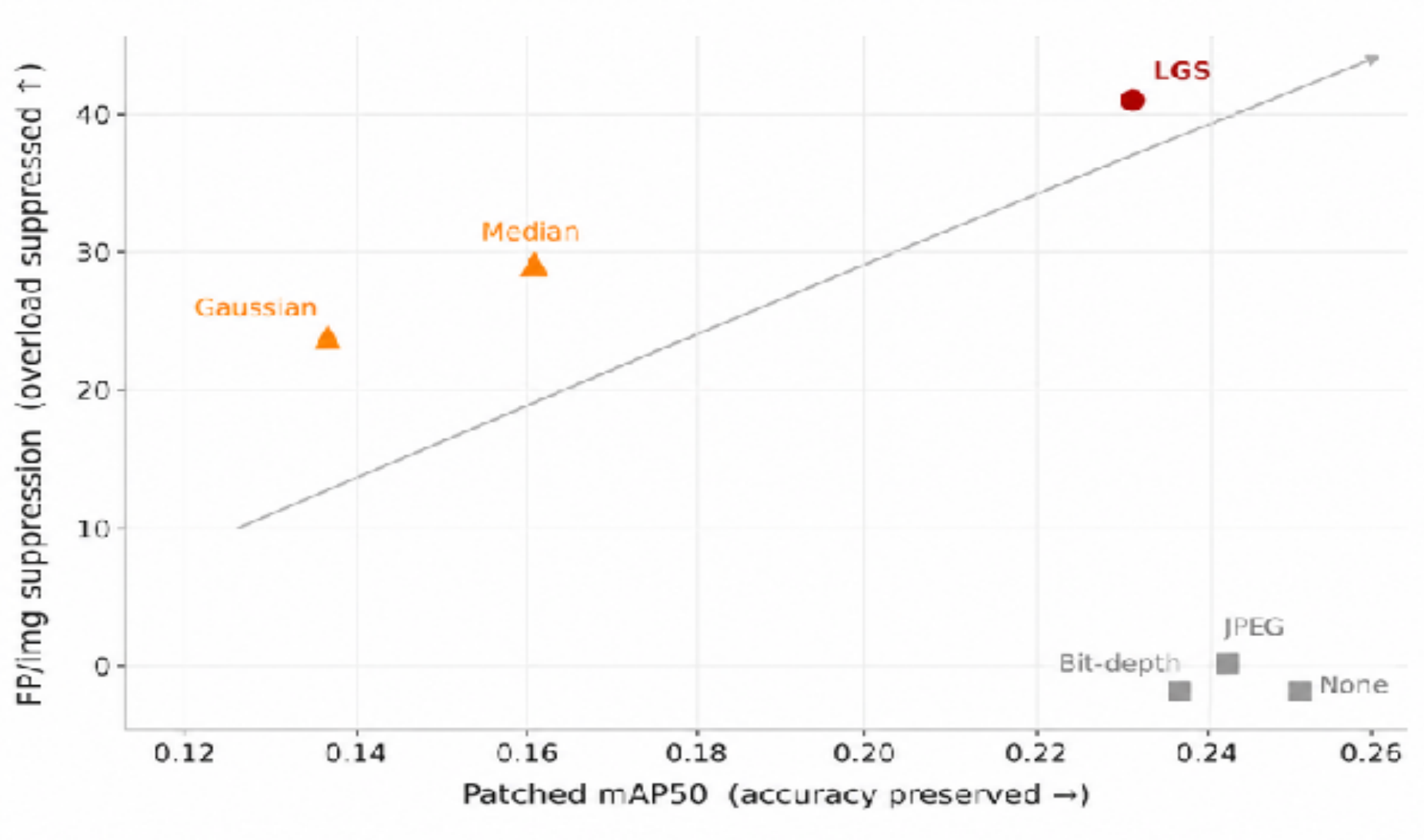
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Smooth locally

Goal: Reduce false positives while preserving detection accuracy

Defense Comparison

Held-out CARLA-GeAR: overload suppression vs retained patched accuracy



LGS Defence in Action



Patched Input



After LGS

Held-out CARLA-GeAR

FP/img (conf ≥ 0.5)

2.84 → 0.41

–85%

mAP50 cost: –0.019

Key Findings

Adversarial patches create perception overload, not only accuracy loss.

Perception overload

mAP50:
0.806 → 0.492

FP/image:
20.9 → 67.2

Detector flooded with false positives.

Robust attack

Cross-scenario transfer

Cross-architecture transfer

Hardware-invisible

Effective defense

Local Gradient Smoothing

FP/image: **2.84 → 0.41**

85% reduction
mAP50 change: **–0.019**

Future Work

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Towards real-world perception-overload attacks and defenses

Physical validation

Print and evaluate physical patches

Different lighting and weather

Camera viewpoint and distance variation

Closed-loop evaluation

Integrate with full autonomous driving stack

Measure planning and control impact

Measure safety-critical behavior

Stronger defenses

Adaptive localized defenses

Real-time optimization

Cross-detector generalization

Developing the next generation of resilient AI algorithms tailored for edge applications

Pioneering secure and trustworthy AI in high-stake domains



THANK YOU! Q & A

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